CSE STUDY HUB

DATA COMMUNICATION & NETWORKING UNIT 5

Application Layer Protocols

SOURCE-ALIGNED CONCISE NOTES

Prepared using the uploaded handwritten class notes for this unit. The explanations retain the concepts and examples visible in those notes, with wording cleaned up for revision.

Topics covered in this unit

- DNS and domain-name resolution
- Email architecture and protocols
- FTP, HTTP and web communication
- Quick revision and exam practice

How to use: Learn the definition, draw the diagram/flow where applicable, practise the numerical method, and finish with the revision questions. If a detail is not present in the uploaded pages, it is not added as an asserted class-note topic.

Exam answer structure

For a 5-mark answer: definition → labelled diagram/formula → working or components → example/application → advantage, limitation or comparison.

UNIT 5 · DNS and domain-name resolution

DNS (Domain Name System) translates a human-readable domain name into an IP address. It allows users to use names such as example.com instead of remembering numeric addresses.

A domain name is hierarchical. Labels are read from the specific host/subdomain toward the top-level domain (TLD), such as .com, .org or .edu. A fully qualified domain name identifies a host/domain in the DNS hierarchy.

In recursive resolution, the DNS resolver takes responsibility for finding the answer and returns it to the client. In iterative resolution, a server may refer the requester to another DNS server that is closer to the answer.

DNS commonly uses port 53. Caching stores recent answers for a period determined by the record's TTL, reducing repeated lookups.

UNIT 5 · Email architecture and protocols

Email delivery involves a mail user agent (MUA), mail transfer agent (MTA) and mail delivery/access components. SMTP is used to send or relay outgoing email between clients and mail servers or between mail servers.

POP3 is used by a client to retrieve messages from a mail server, often by downloading them. IMAP keeps messages managed on the server and synchronizes folders and message state across devices.

SMTP is for sending; POP3 and IMAP are for receiving/accessing. The uploaded notes mention SMTP and POP as the basic sending/receiving protocol pair.

MIME extends email so messages can carry non-plain-text content such as attachments, images and formatted content.

UNIT 5 · FTP, HTTP and web communication

FTP (File Transfer Protocol) supports file transfer between a client and server. Traditional FTP uses separate control and data connections and does not encrypt credentials/data by default.

HTTP is the application protocol used for web request/response communication. A browser sends a request and a web server returns a status line, headers and possibly a response body.

Common HTTP methods include GET (retrieve a resource), POST (submit data), PUT (replace/create a resource), PATCH (partially modify) and DELETE (remove a resource).

HTTPS is HTTP protected by TLS, providing encryption and server authentication when correctly configured. HTTP is typically port 80 and HTTPS port 443.

UNIT 5 · Quick revision and exam practice

Remember the protocol-purpose pairs: DNS—name resolution; SMTP—sending email; POP3/IMAP—accessing received email; FTP—file transfer; HTTP/HTTPS—web communication.

Draw DNS hierarchy from root to TLD to second-level domain to subdomain/host. For email, draw sender MUA → SMTP/MTA → receiver mail server → POP3/IMAP → receiver MUA.

Compare recursive and iterative DNS resolution. Compare POP3 and IMAP. Explain an HTTP request/response cycle and the purpose of HTTPS.

Revision questions: (1) Explain DNS and its resolution methods. (2) Differentiate SMTP, POP3 and IMAP. (3) Explain FTP. (4) Describe HTTP methods and status responses. (5) Differentiate HTTP and HTTPS.